

Calibrated Structured Decisions for LLM-Based Applicant Assessment: An Empirical Study of System One Models in Candidate Ranking

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Abstract—Large language model (LLM) pipelines for automated applicant ranking typically pair free-text assessment generation with iterative listwise comparison, an approach that requires dedicated hallucination-mitigation machinery and offers no native measure of the model’s own uncertainty. In September 2026, TypeSafe AI introduced Jev, the first publicly available “System One Model”: a non-autoregressive architecture trained with Reinforcement Learning for Calibrated Decisions (RLCD) that answers a fixed set of typed questions in a single parallel pass and reports a calibrated confidence with every answer, removing free-text generation – and the hallucination risk it carries – by construction. As of this writing, no independent academic evaluation of this model class exists. This paper’s objective is to close that gap by proposing and empirically validating three architecture-level adaptations that address concrete reliability limitations observed when Jev is applied, unmodified, to applicant assessment: (i) criteria-grounded question design, replacing bare ordinal labels with concrete situational descriptions to improve answer calibration; (ii) multi-call ensemble averaging, aggregating independent single-pass calls to recover ranking stability otherwise lost to single-sample noise; and (iii) structured score decomposition, using the model’s per-requirement granularity as a diagnostic signal for upstream pipeline defects that are invisible to monolithic free-text judgments. The proposed pipeline replaces the free-text assessment and iterative Plackett-Luce tournament stages of a prior LLM-and-tournament-based applicant-ranking system with Jev-based structured decisions, evaluated on a corpus of 501 resumes across 10 job profiles under the same skill-matching shortlist configuration as the prior system. *[Results placeholder: to be finalized upon completion of the full-corpus evaluation and significance testing; a preliminary study on a 338-pair shortlisted subset indicates that criteria-grounded question design raises per-requirement answer confidence from 0.705 to 0.902 (Wilcoxon signed-rank, $p < 0.001$, large effect), and that three-call ensemble averaging recovers ranking convergence from Kendall- $\tau = 0.906$ to 0.945, approaching the 0.957 reported by the closest prior tournament-based system.]*

Index Terms—Applicant Ranking, Calibration, Ensemble Aggregation, Human Resources, Large Language Models, Structured Decision Models, System One Models.

I. INTRODUCTION

RECRUITMENT platforms increasingly rely on large language models (LLMs) to read resumes, summarize applicant fit, and compare candidates against a job description in place of manual screening or keyword search [1]–[4]. These systems earn their usefulness from the same property that also

makes them risky: an LLM generates free text, and free text can assert a strength or weakness that the source resume never supports. Prior work has responded to this hallucination risk with prompt-level constraints, retry-on-contradiction loops, and post-hoc auditing [2], and has combined free-text assessment with iterative listwise tournaments and Plackett-Luce aggregation to turn per-applicant judgments into a defensible ranking [1], [2]. Both classes of mitigation exist because the underlying generation mechanism – next-token sampling over an open vocabulary – offers no structural guarantee against an ungrounded claim and no native signal of how confident the model actually is in what it produced.

A. Research Objectives

This paper’s objective is to determine whether a fundamentally different generation paradigm – one that answers predefined, typed questions instead of generating open text – can replace the free-text assessment stage of an existing applicant-ranking pipeline without reintroducing the reliability problems that paradigm is meant to solve, and, where it does reintroduce new reliability problems of its own, to propose and empirically validate architecture-level adaptations that address them. Concretely, the paper (1) re-implements an existing LLM-and-tournament applicant-ranking pipeline’s assessment and ranking stages around Jev, TypeSafe AI’s first System One Model; (2) identifies, through controlled empirical study on the pipeline’s own corpus, three specific failure modes that appear when Jev is used without adaptation; and (3) proposes and validates a targeted architectural response to each.

B. Gap Analysis

Three gaps motivate this study. First, System One Models are new enough – Jev’s public release predates this manuscript by days – that no independent, peer-reviewed evaluation of the paradigm exists; every technical claim about its reliability currently originates from the vendor itself. Second, the substantial recent literature on LLM calibration and self-consistency [5]–[7] targets autoregressive judges that generate a scalar score or a free-text verdict; none of it addresses a model whose primitive output types are a bounded probability (Noul), a categorical distribution (Choice), or an ordinal distribution (Score) answered in a single parallel pass, so neither its diagnostic

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tools nor its mitigation strategies (e.g., resampling temperature, chain-of-thought prompting) transfer directly. Third, the LLM-hiring-fairness literature [8]–[10] audits demographic bias in a system’s final decision but treats the underlying judgment mechanism as a black box; it does not ask whether the mechanism’s own internal granularity could be used as a diagnostic tool for defects elsewhere in the pipeline that produced the candidates being judged in the first place.

C. Contributions

This paper makes three architecture-level contributions, each grounded in a controlled empirical finding rather than a purely theoretical argument:

- 1) **Criteria-Grounded Question Design:** a methodology for authoring Score-type questions using concrete, situational level descriptions rather than bare ordinal labels, shown by a paired ablation to substantially improve answer confidence on a held-out sample of the same (job, applicant) pairs.
- 2) **Multi-Call Ensemble Averaging:** an aggregation scheme that issues several independent Jev calls per candidate and combines them – arithmetic mean for score-valued answers, confidence-weighted majority vote for categorical answers – shown to recover most of the ranking-order stability of an iterative tournament without an iterative ranking process.
- 3) **Structured Score Decomposition as a Diagnostic Signal:** a use of the model’s per-requirement score granularity, unavailable from a monolithic free-text judgment, to detect and localize a pre-existing upstream shortlisting defect that a prose-only system could not expose without manual inspection.

Section II reviews related work; Section III details the proposed pipeline and its three architectural contributions.

II. LITERATURE REVIEW

A. LLM-Based Applicant Ranking Pipelines

The system this paper extends combines LLM-driven free-text assessment with an active listwise tournament: subsets of shortlisted applicants are repeatedly compared by an LLM, and the resulting partial orderings are aggregated into global utilities via a Bayesian Plackett-Luce model under Monte Carlo knowledge-gradient subset selection, following the design of Yuksel et al. [1]. That prior system further introduced a retry-on-contradiction loop that re-checks every generated weakness against the applicant’s own previously extracted skills, positional-token identifiers to prevent a locally hosted model from mis-copying applicant labels during ranking, and pool-size-adaptive iteration scaling [2]. Other recent systems apply LLMs to resume extraction and pointwise scoring inside retrieval-augmented or multi-agent architectures without a comparative ranking step [3], [4]; none of this body of work departs from free-text or free-form JSON generation as the model’s fundamental output mode, so all of it inherits some version of the hallucination-mitigation burden that motivates [2]’s retry loop.

B. Calibration and Structured Decision Models

A separate and rapidly growing literature studies whether an LLM’s stated or implied confidence reflects its true probability of being correct. Yaldiz et al. [5] show that reinforcement learning from verifiable rewards, while improving task accuracy in decision-making LLMs, produces markedly overconfident models relative to supervised fine-tuning, and propose a calibration-aware reinforcement objective that trades a small amount of accuracy for substantially better-calibrated confidence. Wang et al. [6] propose a structured-output reliability framework built on semantic tree-edit-distance and repeated-measurement consistency scoring, finding that reliability varies sharply across models and prompts even holding the output schema fixed. Both studies, like the broader calibration literature they represent, treat the underlying model as autoregressive: confidence is estimated from output-token probabilities or from resampling a free-text or JSON generation process. Jev’s Noul, Choice, and Score primitives instead report confidence as a first-class field of a single non-autoregressive inference call, trained explicitly for that purpose under RLCD rather than estimated post hoc – a distinct enough mechanism that findings from the autoregressive-calibration literature cannot be assumed to transfer without independent verification, which this paper undertakes for the hiring domain specifically.

C. Self-Consistency and Aggregation for LLM Judgments

Haldar and Hockenmaier [7] document substantial self-inconsistency in LLM-as-a-judge frameworks: the same judge, asked to re-evaluate the same input, frequently returns a different verdict, which undermines any pairwise or ranking comparison built on a single judgment. The prevailing mitigation in that literature is aggregation – majority voting or averaging over multiple independent judgments to suppress pointwise noise rather than eliminating it at the source. This paper adopts the same aggregation principle but applies it to a model whose single call is already an order of magnitude cheaper and faster than the autoregressive judges that motivated the technique, making multi-call aggregation a comparatively inexpensive architectural option rather than a costly last resort.

D. Fairness and Bias in LLM-Based Hiring

A substantial and independent line of work audits demographic bias in LLM-based hiring tools. Webster [8] finds that fairness and competence are separable failure modes: some resume-screening platforms that pass a narrow bias audit fail a resume-role competence test outright, and argues that both must be audited jointly. Wen et al. [9] and Seshadri et al. [10] each document persistent racial and gender disparities in LLM-driven resume evaluation and candidate retrieval, including in stages upstream of any final scoring decision. This literature evaluates a hiring system’s final output against a demographic ground truth; it does not ask whether the judgment model’s own internal structure can be used as an instrument for detecting non-demographic defects – such as an upstream retrieval stage admitting candidates who lack a role’s defining skill – which is the diagnostic use this paper proposes for

Jev’s structured score decomposition. Fairness auditing and pipeline-defect diagnosis are complementary, not competing, concerns, and this paper’s scope is limited to the latter.

E. Positioning of This Work

No paper reviewed here evaluates a System One Model, and no paper in the calibration or self-consistency literature addresses a non-autoregressive, single-pass, typed-question architecture. This paper is, to the authors’ knowledge, the first independent empirical study of this model class, situated in a domain – applicant ranking – already served by a directly comparable free-text predecessor [1], [2], which makes an architecture-level, rather than purely benchmark-level, comparison possible.

III. METHODOLOGY

A. Pipeline Overview

The proposed pipeline retains the ingestion, LLM-based skill extraction, and embedding-based shortlisting stages of the prior system [2] unmodified: job profiles are loaded as plain text and resumes are parsed from PDF; an LLM extracts a normalized, resume-grounded skill list for each job profile and applicant; every extracted skill is embedded into a shared matrix; and, per job profile, an applicant is shortlisted only if at least a minimum number of the profile’s distinct technical skills are matched above a cosine-similarity threshold. The pipeline departs from the prior system at the next two stages. Where the prior system generated a free-text assessment (strengths, weaknesses, additional skills) and then ranked shortlisted applicants through an iterative Monte Carlo knowledge-gradient tournament with Plackett-Luce aggregation, the proposed pipeline issues one or more Jev calls per shortlisted (job, applicant) pair, each returning a structured assessment directly, and ranks applicants for a job profile by sorting on the resulting overall fit score – removing the iterative ranking stage entirely.

B. Structured Assessment via Jev

Each Jev call receives a single textual state – the job title, job description, the applicant’s previously extracted skills, and the applicant’s resume text – together with a fixed set of typed questions answered against that state in one parallel pass. Three questions are asked of every (job, applicant) pair regardless of job profile: a Score question estimating overall fit on a five-level scale, a Choice question selecting a hiring recommendation among “hire,” “maybe,” and “no,” and a Noul question estimating the probability that the applicant meets the job profile’s stated minimum qualifications. One additional Score question is generated per distinct technical skill in the job profile’s extracted requirement list, asking how well the applicant’s resume evidences that specific requirement. The job-description extraction stage additionally identifies, where stated, named certifications, a seniority/experience requirement, and an education requirement; each yields one further Noul question (one per certification, one for seniority, one for education), asked only when the job description states

that requirement, so that a job profile with no certification or seniority language incurs no extra calls. This decomposes the prior system’s single, opaque “meets minimum qualifications” judgment into diagnosable sub-components – a candidate failing due to a missing certification is now distinguishable from one failing due to insufficient experience – while the original holistic question is retained unchanged alongside them, mirroring the relationship between the overall fit score and the per-requirement scores. Every answer carries a model-reported confidence, and Score-type answers additionally carry a probability distribution over the five levels; the reported score is mapped linearly onto a 0–100 scale for downstream ranking and reporting.

C. Novelty 1: Criteria-Grounded Question Design

An initial implementation defined every Score question’s five levels with bare numeric labels (“0,” “25,” “50,” “75,” “100”), a design later found to yield markedly lower answer confidence than a revised design using concrete, situational level descriptions – for example, distinguishing a requirement “mentioned only as a bare skill-list item” from one “used substantively in a real role or project with clear responsibility.” To isolate the effect of this change from ordinary sampling noise, the two designs were compared in a paired ablation: for a sampled subset of (job, applicant) pairs already assessed under the concrete-criteria design, a second call was issued using the original bare-label design against the identical state, and answer confidence was compared pairwise across every shared question, separating the one criteria-affected overall-fit question, the criteria-unaffected recommendation and minimum-qualification questions (a negative control, since their criteria were never changed), and the many per-requirement questions (the primary criteria-affected quantity) into three distinct comparisons. A Wilcoxon signed-rank test with a rank-biserial effect size was applied to each comparison.

D. Novelty 2: Multi-Call Ensemble Averaging

Because each Jev call is independent and comparatively inexpensive, the pipeline issues n independent calls per (job, applicant) pair – three by default, matching the stability-repeat count of the prior tournament-based system [2] – and aggregates them into a single assessment. Numeric fields (overall fit score, per-requirement scores, and per-question confidence) are combined by arithmetic mean across the n calls. The categorical recommendation is combined by majority vote, with ties broken in favor of the tied label carrying the higher mean confidence across the calls that selected it. Binary answers (minimum-qualification, certification, seniority, and education judgments) are combined by simple majority, defaulting to the conservative (negative) outcome on an exact tie. To evaluate whether this aggregation actually improves ranking-order stability rather than merely averaging away genuine disagreement, ranking convergence was measured two ways on the same corpus: first from single, unaggregated calls (three independent per-pair scores, each ranked separately, with pairwise Kendall- τ computed across the resulting three rankings per job profile), and second from three independent

three-call averages (nine total calls per pair, grouped into three trials of three, each trial’s ranking built from its own three-call average, with pairwise Kendall- τ again computed across the three trial rankings).

E. Novelty 3: Structured Score Decomposition as a Diagnostic Signal

Because Jev reports a separate score for every extracted requirement rather than a single free-text verdict, a job profile’s shortlisted applicant pool can be inspected at the granularity of individual skills rather than only at the level of an aggregate score. This paper uses that granularity diagnostically: for each job profile, the signal-to-noise ratio of its shortlisted applicant pool – the standard deviation of applicants’ mean overall-fit scores (between-applicant signal) divided by the mean within-applicant standard deviation across repeated calls (measurement noise) – is correlated (Spearman) against that profile’s ranking-convergence Kendall- τ , to test whether ranking instability traces to genuine score-noise proximity between similarly qualified applicants rather than to model unreliability. Independently, for job profiles exhibiting anomalously low overall scores, the lowest-scoring shortlisted applicants’ full per-requirement breakdowns are inspected directly to determine whether low scores reflect genuinely weak applicants or a shortlisting-stage admission error – for example, an applicant shortlisted for a role through generic, widely shared skills while scoring zero on every skill that specifically defines that role.

F. Experimental Setup

The pipeline is evaluated on the same corpus and shortlist configuration as the prior system [2]: 501 resumes across 10 synthetically authored job profiles, shortlisted under a cosine-similarity threshold of 0.8 and a minimum of five matched distinct technical skills per job profile. Four evaluation procedures are applied to the resulting shortlisted (job, applicant) pairs: test-retest reliability, issuing three independent single calls per pair and measuring score variance and recommendation agreement across them; internal coherence, testing whether the model’s own structured outputs agree with one another (per-requirement scores against the overall fit score via Spearman correlation; the categorical recommendation and the binary minimum-qualification answer against the overall fit score via Mann-Whitney U); ranking convergence, the Kendall- τ procedure described in Section III-D, computed both from single calls and from three-call averages; and the criteria-design ablation described in Section III-C. All statistical tests are two-sided except where a directional hypothesis (e.g., that a “hire” recommendation carries a higher score than a “no” recommendation) is explicitly stated, and all reported effect sizes accompany, rather than substitute for, the corresponding significance test.

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